# Air Quality

Pupil sheet

**Name:** _____ **Date:** _____

**Safety:** Do not move the boxes unless told to; the teacher opens the window; never climb on the sill.

## 1. What I see on the screen

The screen shows **one gauge per box** and a summary line: $\Delta$**CO₂**, the difference between the highest box and the lowest. With only one box it reads **Connect at least two boxes**. Each box has its own name and colour, the same on the gauge and on the chart.

**Under the Δ line the app writes two lines.** The **Level** comes from the highest value: from 1200 ppm "the air is getting stale", from 2000 ppm "open the windows now". The **Difference** uses *fixed thresholds* — 300 and 600 ppm — which know nothing about our boxes. **I check it against my own threshold.**

**Our detection threshold** (from co-location) = _____ **ppm**
A difference smaller than that *cannot be told* from noise.

## 2. What I do on my own

**1. Co-location.** Every box on the same table, ten minutes. I record $\Delta CO_2$ over the last five minutes; the largest value is the threshold above.

| **Minute** | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|
| Δ (ppm) | | | | | |

**First I predict.** The corner furthest from the window will be about _____ ppm higher than the spot beside it.

because _____

_____

**2. Every location, at the same moment.** Time: _____ How many of us: _____ Window: ☐ shut ☐ open

| Location | Box | $CO_2$ (ppm) | Band (level) |
|---|---|---|---|
| Far corner | | | |
| By the window | | | |
| By the door | | | |
| Middle of the room | | | |

**3. I compare every pair with my threshold. "Borderline"** is a correct answer, not a dodge.

| The pair | Δ (ppm) | Verdict (circle one) |
|---|---|---|
| Far corner vs window | | detected / borderline / not detected |
| Far corner vs door | | detected / borderline / not detected |
| Middle vs window | | detected / borderline / not detected |

**4. Airing.** We open the window. $\Delta CO_2$ immediately afterwards: _____ ppm. Did it rise? ☐ yes ☐ no
After ten minutes: _____ ppm.

## 3. The conclusion I write

I predicted that, and the data show that _____

_____

When we opened the window, Δ rose because _____

_____

The location most in need of air is _____, because _____

_____

**Exit ticket**

What is our detection threshold, and how do we know it?

_____